

Name: Sofia Isabela Castro Cáceres		
School: Academica Británica Cuscatleca		
Group: 4th seg...		Sex: Female
Date of first test: 13/11/2024	Level: 8B	Age: 9:01
Date of second test: 10/10/2025	Level: 9B	Age: 10:00

Scores

No. attempted (/54)	SAS	SAS (with 90% confidence bands)										Overall ST	PR	GR (/25)	End of KS2 indicator	Progress Category
		60	70	80	90	100	110	120	130	140						
48	83										3	13	16	95	Expected	

Previous & Current Scores

Date	Age at test (yrs:mths)	Level / Form	No. attempted	SAS	SAS (with 90% confidence bands)										ST	PR	End of KS2 indicator
					60	70	80	90	100	110	120	130	140				
07/05/2025	9:07	9A	40 / 54	72											1	3	87
10/10/2025	10:00	9B	48 / 54	83											3	13	95

Analysis of Curriculum content categories

Curriculum content category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Number	38	39%	62%	-23%
Measurement	6	25%	47%	-22%
Geometry	4	17%	40%	-23%
Statistics	6	33%	63%	-30%

Analysis of Process categories

Process category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Fluency in facts and procedures	13	31%	64%	-33%
Fluency in conceptual understanding	22	50%	64%	-14%
Problem solving	7	33%	34%	-1%
Mathematical reasoning	12	13%	57%	-44%

Implications for teaching and learning

- By comparing scores from a previous administration of *PTM* it is possible to categorise progress as expected, higher or lower than expected or much higher or much lower than expected.
 - Sofia Isabela took *PTM8B* in November 2024 and from then until now has made expected progress in maths.
- Reviewing the *Analysis of Curriculum content categories* will help to identify where there are specific strengths and weaknesses and to plan next steps.
- When performance in a test like *PTM* is uneven, Sofia Isabela's performance should be investigated further, especially when performance in part of the test is low. If the school uses CAT this may offer a further insight into Sofia Isabela's learning profile, which will inform the diagnosis.
- Sofia Isabela appears to have a barrier to working on mental maths that is not there for written maths tasks. Specific structured support will be necessary in order for Sofia Isabela to be able to overcome this discrepancy. It may be merely a lack of practice with mental calculations, or an over-reliance on taught methodologies together with a reluctance to experiment and try out different approaches. It is this experimentation and an exposure to practical applications that develop confidence and lead to the mental agility in the manipulation of numbers that lies at the heart of mental maths.
- Where scores are fairly evenly balanced across the curriculum categories, this suggests that Sofia Isabela will generally demonstrate a level of understanding of mathematical concepts appropriate for this age, irrespective of whether this is in a format requiring steps in a calculation to be written down, or in mental maths when only 'jottings' or nothing is written. Sofia Isabela is developing the language of mathematics broadly in line with expectations for her age group. Fluency and agility are better developed in applying and understanding maths than mental maths.
- Where scores across the curriculum categories are uneven, specific areas of weakness might be addressed as follows:
 - Further targeted practice in the areas identified as being relatively weaker.
 - Practical activities using equipment that is designed to help Sofia Isabela to 'see' the thinking that lies behind any concepts that are not yet secure.
 - Get Sofia Isabela to explain workings to another pupil so that any misconceptions can be highlighted and corrected through discussion.
- Check that language is not a barrier by using suitable but varied vocabulary; continue to use practical apparatus and keep the interventions either one-to-one or in a very small group in order to build confidence. Make time every day to practise these skills to develop confidence. Regularly provide suitable equipment for Sofia Isabela to manipulate and explore how and why a calculation strategy works, which will help Sofia Isabela to describe and visualise or 'see' the method working. The equipment can include objects like counters, interlocking cubes, coins, counting sticks, bead strings, number lines, 100-squares, place-value cards, structural apparatus like Dienes blocks, diagrams of shapes divided into fractional parts, home-made packs of cards with numbers and operators on and so on.
- An interactive whiteboard can also be a powerful tool for manipulating images.
- Offer Sofia Isabela a chance to practise similar calculations in appropriate ways without recording these thought processes on paper. Articulating and vocalising the steps in a calculation is the precursor to being able to carry it out without writing it down. Practical activities that give rise to a dialogue about numbers, shapes and measurements further supports the development of mathematical language and the ability to perform calculations without recourse to formal recording.
- Make sure that Sofia Isabela begins to use the written language for mathematics that she is developing to aid mental agility. The setting out of written calculations in the correct way is fundamental to success and can aid understanding. For example, an understanding of place value is consolidated by setting out calculations in columns. Correct setting out can be used to explore with Sofia Isabela ways of undertaking these calculations without writing down the whole of the sum. This can increase confidence and support progress in mental maths.

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Analysis by question

The table below shows each question in the test and the pupil's response (correct/incorrect) compared with the group and standardisation average.

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
MM1	Number	Fluency in facts and procedures	What is six doubled?	0 / 1	72	87
MM2	Number	Fluency in facts and procedures	Multiply three by seven.	0 / 1	56	63
MM3	Number	Fluency in conceptual understanding	Write in figures the number four thousand and thirty-six.	0 / 1	48	63
MM4	Number	Mathematical reasoning	A packet of forty-five sweets is shared equally between five people. How many sweets does each person get?	0 / 1	20	64
MM5	Number	Fluency in facts and procedures	A cake is cut into thirds. How many pieces are there?	0 / 1	68	56
MM6	Measures	Fluency in conceptual understanding	My watch says half past two in the afternoon. What would a twenty four hour digital watch say?	0 / 1	4	28
MM7	Number	Mathematical reasoning	One car has four tyres. How many tyres do three cars have?	0 / 1	60	80
MM8	Number	Mathematical reasoning	There are six eggs in one box. How many eggs are there in seven boxes?	0 / 1	12	52
MM9	Number	Fluency in conceptual understanding	What is twenty-five take away thirteen?	0 / 1	48	66
MM10	Number	Fluency in facts and procedures	Write a multiple of six that is between fifteen and twenty-five.	0 / 1	20	44
MM11	Number	Fluency in conceptual understanding	Look at these numbers. Write down all the odd numbers.	0 / 1	48	64
MM12	Number	Fluency in facts and procedures	Add these numbers: forty-one, twenty-five and fourteen.	0 / 1	16	43
MM13	Number	Mathematical reasoning	There are thirty-two people in a restaurant. Thirteen are adults. How many are children?	0 / 1	12	36
MM14	Measures	Mathematical reasoning	I have twenty-seven pence. I want to spend thirty-five pence. How much more money do I need?	0 / 1	48	61
MM15	Measures	Fluency in facts and procedures	How many days are there in eight weeks?	0 / 1	0	47
AU1	Number	Fluency in conceptual understanding	Draw lines to connect the matching fractions.	2 / 2	90	89
AU2	Number	Fluency in conceptual understanding	What fraction of this shape is shaded?	1 / 1	40	78
AU3a	Number	Fluency in conceptual understanding	Which day had the lowest midnight temperature?	0 / 1	48	59

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU3b	Number	Fluency in conceptual understanding	Which day had a midnight temperature 1 degree higher than Friday?	1 / 1	40	50
AU4	Number	Mathematical reasoning	On Sunday the midnight temperature was -2 degrees. The next day, the midnight temperature was 5 degrees higher. What was it?	0 / 1	36	49
AU5	Geometry	Fluency in facts and procedures	Name the 3 shapes that are labelled.	0 / 1	40	46
AU6a	Geometry	Fluency in facts and procedures	In which labelled shape are all the angles right angles?	0 / 1	40	60
AU6b	Geometry	Fluency in facts and procedures	In which labelled shape are all the angles greater than a right angle?	0 / 1	12	42
AU7	Number	Mathematical reasoning	Here are 3 more fraction sums that make 1 whole. Fill in the missing numbers.	0 / 2	14	62
AU8	Measures	Mathematical reasoning	Write the times in the correct boxes.	0 / 2	32	36
AU9	Statistics	Fluency in conceptual understanding	Put a circle round the country that has the least tigers.	1 / 1	80	83
AU10	Statistics	Fluency in conceptual understanding	Roughly how many tigers are there in Malaysia?	1 / 1	76	81
AU11	Statistics	Fluency in conceptual understanding	Put a circle round the 3 countries that have less than 500 tigers.	0 / 1	64	70
AU12a	Number	Fluency in conceptual understanding	Which number has 7 as its hundreds digit?	1 / 1	72	83
AU12b	Number	Fluency in conceptual understanding	Which number is 10 times bigger than 62?	1 / 1	24	63
AU12c	Number	Fluency in conceptual understanding	Which number is not a whole number?	1 / 1	56	70
AU12d	Number	Fluency in conceptual understanding	Which is the smallest number?	1 / 1	68	81
AU13	Number	Fluency in facts and procedures	Here is a number machine. Fill in the missing number.	1 / 1	84	74
AU14a	Number	Mathematical reasoning	Fill in the missing number.	1 / 1	64	66
AU14b	Number	Problem solving	What number comes out the same as it goes in?	1 / 1	44	40
AU15a	Number	Fluency in facts and procedures	Which is the biggest number?	1 / 1	96	94
AU15b	Number	Fluency in facts and procedures	Write in a number that is bigger than the biggest number here.	1 / 1	92	86
AU16a	Number	Fluency in facts and procedures	Which are the smallest two numbers in the list?	1 / 1	96	93
AU16b	Number	Mathematical reasoning	Write in any whole number that is between the 2 smallest numbers.	0 / 1	52	74
AU17	Measures	Fluency in conceptual understanding	Which two of these are equal to seventy pence?	1 / 2	44	68
AU18	Number	Fluency in conceptual understanding	What number is exactly between 41 and 55?	0 / 1	4	31
AU19	Number	Fluency in conceptual understanding	What number is exactly between 4.5 and 5.5?	0 / 1	24	51
AU20	Number	Fluency in conceptual understanding	What number is exactly halfway between $\frac{2}{7}$ and $\frac{4}{7}$?	0 / 1	40	71

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU21	Geometry	Problem solving	Write the letters of each shape into the correct boxes.	1 / 3	13	31
AU22	Number	Mathematical reasoning	Now choose 4 different numbers to show more multiplications with the answer 32.	0 / 2	20	47
AU23	Measures	Fluency in conceptual understanding	At the bank, Mr Smith changes a £10 note for 20p pieces. How many 20p pieces does he get?	1 / 1	12	29
AU24	Number	Mathematical reasoning	What is the smallest possible number you can make using all 4 of these cards?	1 / 1	72	84
AU25	Number	Problem solving	Arrange the cards to make 2 numbers whose sum is 74.	0 / 1	52	57
AU26	Number	Problem solving	Arrange the cards to make 2 numbers whose difference is 26.	1 / 1	24	35
AU27a	Number	Problem solving	How many stickers will be left when she has made all the cards?	0 / 1	20	21
AU27b	Number	Problem solving	How many more invitations can she make with the left-over stickers?	0 / 1	12	24
AU28	Statistics	Fluency in conceptual understanding	How many babies were at the medical centre on that Wednesday?	0 / 1	20	40
AU29	Statistics	Fluency in conceptual understanding	Put an x on the graph in question 17 to show John's heartbeat.	0 / 1	64	70
AU30	Statistics	Problem solving	Which 2 of these age groups could Erin be in?	0 / 1	20	33